PROJECT REPORT

App Insights Unlocked

Google Play Store Data Analytics Using Power BI

A business-focused analysis of app performance, ratings, installs, pricing, app size, and customer sentiment

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Tools: Microsoft Power BI • Power Query • DAX

Note: This report defines the analytical approach and intended deliverables. Findings and KPI values must be populated after the cleaned data is analyzed in Power BI.

Executive Summary

This project examines Google Play Store app data to understand the factors associated with app performance and user satisfaction. The analysis uses two datasets: an app-level dataset containing details such as category, rating, reviews, installs, size, price, and update date; and a user-review dataset containing review text and sentiment information. Power BI is used to clean, model, analyze, and present the information in an interactive report.

The dashboard is designed to help app developers, product managers, marketing teams, and senior management compare category performance, explore the relationship between installs and ratings, review free-versus-paid app patterns, and investigate positive and negative user feedback. The report separates planned analysis from confirmed findings: numerical conclusions should only be added after the data has been cleaned, measures validated, and visuals reviewed.

1. Introduction

Mobile app marketplaces contain a wide range of applications with different audiences, business models, sizes, and levels of user engagement. App ratings and review activity can offer useful signals about user experience, while install figures can help indicate reported reach. Looking at these measures together provides a more balanced view than relying on a single metric.

The purpose of this project is to turn raw Google Play Store data into clear, practical insights. The work focuses on describing market patterns and relationships in the supplied data. It does not assume that correlation proves causation or that the dataset represents every app or every review on Google Play.

2. Problem Statement

The business needs a structured way to understand what distinguishes better-performing apps, which categories attract greater interest, and how ratings, app size, pricing, installs, and customer sentiment vary across the available records. The analysis should support product and marketing decisions by highlighting patterns that deserve further investigation.

3. Project Objectives

- Measure the number of apps, average rating, reported installs, and review counts.
- Compare app counts and average ratings across categories and content-rating groups.
- Examine differences between free and paid apps.
- Explore the relationship between app size, price, installs, and ratings.
- Summarize positive, negative, and neutral sentiment in the available review records.
- Present the analysis in an interactive Power BI dashboard and translate validated results into practical recommendations.

4. Stakeholders and Business Value

Stakeholder	Potential value from the analysis
App developers	Identify app categories and user-feedback themes that may guide product improvements.
Product managers	Compare ratings, app size, pricing, and adoption indicators when prioritizing product work.
Marketing team	Understand category-level reported demand and explore differences between free and paid apps.
Senior management	Review a concise performance overview and use the evidence to prioritize follow-up analysis.
App users and partners	Indirectly benefit when product teams use feedback to improve app quality and relevance.

5. Data Sources and Description

The project uses the two supplied CSV files and the project brief, "App Insights Unlocked: A Data Analytics Challenge." The brief identifies app ratings, reviews, size, installs, type, price, content rating, genres, and update date as key analysis fields.

Dataset	Grain / purpose	Important fields
googleplaystore.csv	App-level records; used for marketplace and performance analysis.	App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated
googleplaystore_user_reviews.csv	Review-level records; used for sentiment and review-text analysis.	App, Translated_Review, Sentiment, Sentiment_Polarity, Sentiment_Subjectivity

The two datasets have different levels of detail. One app may be associated with many review records, and the review file may not contain a matching review for every app. These differences must be considered during data modeling and interpretation.

6. Tools and Methodology

- Power BI Desktop: data model, DAX measures, visuals, slicers, and report pages.
- Power Query: data profiling, missing-value checks, text standardization, duplicate investigation, and data-type conversion.
- DAX: reusable measures for app counts, ratings, installs, reviews, and sentiment percentages.
- Exploratory analysis: category comparisons, install bands, price comparisons, app-size comparisons, and review sentiment breakdowns.

7. Data Cleaning and Preparation

The following checks should be completed and documented before interpreting dashboard results:

1. Review missing values in Rating, Size, Type, and other analysis fields. A missing rating should not be replaced with zero.
2. Convert Reviews to a numeric type and clean Installs by removing display characters such as commas and plus signs.

3. Convert Price to a numeric value after removing currency symbols.
4. Standardize Size values to megabytes, converting kilobytes where necessary and treating “Varies with device” as unknown.

5. Convert Last Updated to a valid date type.
6. Standardize text fields such as Category, Type, Genres, and Sentiment.
7. Investigate duplicate app records. Do not remove repeated app names automatically unless the records have been checked.
8. Remove blank review text only for analyses that require review text; retain other valid records when useful.
9. Check unmatched app names between the app dataset and the review dataset.

Install values in the source are expressed as thresholds such as “1,000+” and “10,000+.” Removing formatting characters makes them numeric, but does not turn the reported thresholds into exact install counts. Any summed install metric should be described carefully and should be calculated only after duplicate records have been assessed.

8. Data Model

A recommended model uses a distinct app dimension table, DimApp, containing one row per app name. DimApp[App] connects to GooglePlayStore[App] and UserReviews[App] using one-to-many relationships with single-direction filtering from the dimension to the two data tables. This avoids directly joining every app record to every review record, which could multiply app-level values.

Before creating relationships, confirm that app names are unique in DimApp, the key columns use compatible data types, and unmatched names are understood. If the source contains conflicting records for the same app, document the rule used to select or aggregate them.

9. Key Performance Indicators

KPI	Definition	Interpretation / caution
Total Apps	Distinct count of app names.	Check duplicate and conflicting app records.
Average Rating	Average of nonblank app ratings.	Compare alongside the number of rated apps.
Reported Installs	Sum of cleaned install values, if appropriate.	Source values are thresholds; do not describe them as exact user counts.
App Review Count	Sum of app-level review counts, if appropriate.	Validate duplicate app records and distinguish from review-file row count.
Free / Paid Apps	Distinct app counts by Type.	Investigate missing or invalid Type values.
Average App Size	Average of valid Size_MB values.	Unknown or variable sizes should remain blank.
Review Record Count	Number of review records after cleaning.	This is the supplied review sample, not necessarily all reviews.
Positive Review %	Positive sentiment records divided by classified review records or the chosen review denominator.	State the denominator and treatment of missing sentiment.

10. Dashboard Design

Page 1 — App Performance Overview

This page gives stakeholders a quick summary of the app dataset. Recommended visuals include KPI cards for Total Apps, Average Rating, Reported Installs, App Review Count, Free Apps, and Paid Apps; a bar chart of app counts by category; a chart of average rating by category; a free-versus-paid distribution; and a content-rating distribution.

Primary question: What does the overall app marketplace look like, and how do categories differ?

Page 2 — Factors Associated with App Performance

This page explores the relationship between installs and ratings, app size and installs, and price and ratings for paid apps. Recommended visuals include scatter charts, average rating by install band, a Top 10 app table, and slicers for category, type, content rating, and install band. Use tooltips to display app name, category, rating, reviews, installs, and price.

These visuals show associations rather than proving that price, size, or installs cause a particular rating or outcome. Consider the effect of category mix, missing values, outliers, and the source's install thresholds.

Page 3 — Customer Sentiment Analysis

This page summarizes positive, negative, and neutral sentiment in the review dataset. Recommended visuals include sentiment KPI cards, a sentiment distribution chart, sentiment counts by category, a positive-versus-negative comparison, and a review-detail table with review text and sentiment scores. Use category and sentiment slicers to explore the data.

Review themes such as performance problems, crashes, navigation, or advertising should be reported only if they are observed in the actual review text. A high count of negative reviews can reflect greater review volume, so percentages and denominators should also be considered.

11. Analysis Questions

10. What is the average rating across apps with available ratings?
11. How many distinct categories and apps are represented?
12. How are app sizes distributed, and how do sizes vary by category?
13. What is the distribution of free and paid apps?
14. Which apps have the highest reported install thresholds?
15. How many apps have ratings of 4.0 or above?
16. How do average review counts differ between free and paid apps?
17. Which categories have the highest average ratings, and how many rated apps support each result?
18. How do ratings vary across install bands?
19. How do paid-app ratings vary by price range?
20. How do app size and installs vary together?
21. What is the distribution of sentiment, and which review themes appear repeatedly in negative feedback?
22. How many apps were last updated in 2018, and what patterns appear in update dates?

12. Findings and Results

This section should be completed after the cleaned data has been analyzed in Power BI. The supplied project brief defines the questions to investigate, but it does not provide verified answers for the supplied CSV files. To avoid presenting unsupported conclusions, no category rankings, average ratings, install totals, or sentiment percentages are invented in this report.

Finding to complete	Evidence to record	Business meaning
Leading categories	Top categories by app count and reported installs.	Potential areas of demand and competition.
Rating patterns	Average rating by category with rated-app counts.	Potential quality benchmarks.
Free vs paid	App counts, ratings, and review-count comparison.	Evidence to inform pricing and positioning.
Size vs installs	Scatter or install-band comparison.	Whether size deserves further product investigation.
Review sentiment	Sentiment percentages and examples of recurring themes.	Potential priorities for user-experience improvements.

13. Recommendations

The recommendations below are proposed decision rules. Final recommendations should be refined using the measured results.

- Prioritize categories only after comparing both reported demand and competition; a large category may also be highly competitive.
- Use average ratings alongside sample size and review volume before treating a category as a quality benchmark.
- Evaluate free and paid apps within comparable categories where possible, rather than assuming pricing alone explains rating differences.
- Investigate whether app size creates a practical user-experience concern before recommending size reduction.
- Review negative comments for recurring themes and validate them with product teams before prioritizing fixes.
- Document data limitations and refresh the analysis when newer or more complete marketplace data becomes available.

14. Limitations

- The analysis is limited to the records in the supplied datasets and may not represent the current Google Play Store.
- Install values are reported as thresholds and are not exact counts.
- Some apps may have missing ratings, sizes, or other fields.
- Duplicate or inconsistent app records can affect app counts and aggregated metrics unless resolved.
- App names may not match perfectly between the two datasets, limiting sentiment coverage.
- Sentiment labels and scores are supplied data fields; their classification quality has not been independently validated in this report.
- Observed relationships are descriptive and do not establish causation.

15. Conclusion

The App Insights Unlocked project provides a structured way to explore app performance and user feedback using Power BI. By combining careful data preparation, a model that respects the different levels of detail in the datasets, reusable DAX measures, and interactive visuals, the report can help stakeholders compare categories and identify areas that merit further investigation.

The value of the project will come from connecting each chart to a clear business question and supporting every conclusion with a validated measure. Once the dashboard is populated, the final report should be updated with the actual findings, the relevant screenshots, and specific recommendations based on the evidence.

Appendix A — Suggested Deliverables

- Power BI report file (.pbix) with three report pages.
- Cleaned and documented data transformation steps in Power Query.
- DAX measures for core KPIs and sentiment analysis.
- Screenshots of each dashboard page.
- Final findings table with measured values and evidence-based recommendations.

Appendix B — Source Material

Primary project source: “App Insights Unlocked: A Data Analytics Challenge” (user-provided PDF). Datasets supplied for the project: googleplaystore.csv and googleplaystore_user_reviews.csv.

Microsoft documentation for implementation: Power Query (https://learn.microsoft.com/en-us/power-query/), Power BI (https://learn.microsoft.com/en-us/power-bi/), and DAX (https://learn.microsoft.com/en-us/dax/).